Anindya Mondal

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Research Areas

Machine Learning, Graph Signal Processing, Computer Vision, Graph Neural Networks, Internet of Things

Education

- **Bachelor of Engineering in Electronics and Telecommunication (with Honours, CGPA 8.79/10)** Aug 2018 - Jun 2022
  Jadavpur University, Kolkata, India
  Bachelor Project: *Recovery of Missing Data in IoT Sensor Networks using Graph Signal Processing*

Relevant Experience

Research Assistant/ Intern

- IACV Lab, Indian Institute of Science, Bengaluru, India — Apr 2022 - Jul 2022
- Texas Instruments Innovation Lab, Jadavpur University, Kolkata, India — Aug 2021 - May 2022
- Imaging, Vision and Pattern Recognition (IVPR) Group, Jadavpur University, India
  & Mathématiques, Image, et Applications (MIA) Lab, La Rochelle Université, France — Oct 2020 - Aug 2021

Trainee

- Indian Institute of Remote Sensing, Indian Space Research Organization, India — Jun 2020 - Oct 2020

Projects

- **Source-free Domain Adaptation**

  Working with Prof. Soma Biswas

  Currently using source-free domain adaptation to perform pixel-level tasks in computer vision.

- **Reconstructing Time-varying Graph Signals using Graph Neural Networks**

  Working with Jhony H. Giraldo and Prof. Thierry BOUWMANS

  Currently working on designing a Graph Convolutional Network (GCN) framework to reconstruct time-varying graph signals from their samples.

- **Recovery of Missing Data in IoT Sensor Networks using Graph Signal Processing**

  Advised by Prof. P. Venkateswaran (Bachelor Project - Jadavpur University)

  Applied a Sobolev norm minimization method for reconstructing time-varying graph signals to recover missing data in sensor networks. Our approach improved upon the previous works by as much as 54%. Paper accepted at **EUSIPCO 2022**.

- **Graph-based Approaches for Moving Object Detection and Semantic Segmentation for Event-based Vision**

  Worked with Prof. Ananda S. Chowdhury, Thierry BOUWMANS, and Jhony H. Giraldo

  Worked on developing a new algorithm based on Graph Spectral Clustering to detect moving objects in a scene captured by an event-based camera and presented the results at the **ICCV Workshops 2021**

  Worked on developing a graph-based semi-supervised learning algorithm for semantic segmentation in event-based data.

Publications & Preprints

- **Anindya Mondal**, Mayukhmali Das, Aditi Chatterjee, Palaniandavar Venkateswaran "**Recovery of Missing Sensor Data by Reconstructing Time-varying Graph Signals**," Accepted in *EUSIPCO - 2022, Belgrade, Serbia* [Preprint] [Presentation Slides]
- **Anindya Mondal**, Shashant R., Jhony H. Giraldo, Thierry Bouwmans, and Ananda S. Chowdhury, "**Moving Object Detection for Event-based Vision using Graph Spectral Clustering**," in *IEEE/CVF International Conference on Computer Vision (ICCV) Workshops -2021, Montreal, BC, Canada* [Preprint] [Code] [DOI] [Presentation Video]
- **Anindya Mondal**, Mayukhmali Das, "**Moving Object Detection for Event-based Vision using k-Means Clustering**," in *IEEE UPCON - 2021, India* [Preprint] [DOI]

Programming Skills

- **Languages**: Python, C, Matlab/ Octave
- **Libraries**: Matplotlib, Pandas, Sklearn, Numpy, PyTorch, PyGSP
- **Applications/ Softwares** : LaTeX, Git
- **Miscellaneous**: Photoshop/ GIMP, Blender, HTML, TSpice, Multisim NI

Awards

- **ACM SIGKDD India Chapter Uplink Research Internship Award**, A stipend of 50000 INR and travel grant for carrying out research work at IACV Lab, IISc Bengaluru 2022
- **Dr. B.R. Ambedkar Talent Award**, Awarded by the Government of West Bengal, India for meritorious performance in school final examination 2016

Relevant Coursework

- **Undergraduate:** Digital Signal Processing, Digital Image Processing, Signals and Systems, Data Structures and Algorithms, Embedded Systems, Engineering Mathematics, Control Systems, Operating Systems and System Software, Digital Switching and Computer Networks, Analog and Digital Systems, VLSI Design, Physical Electronics and Devices

Selected MOOCS and Certifications

- **Machine Learning** | Stanford University | Coursera
- **CS224W: Machine Learning with Graphs** | Stanford University
- **Python Programming** | University of Michigan | Coursera
- **Neural Networks and Deep Learning** | deeplearning.ai | Coursera
- **Computer Vision Basics** | University at Buffalo and SUNY | Coursera
- **Introduction to Programming with MATLAB** | Vanderbilt University | Coursera
- **Digital Signal Processing Specialization** | EPFL | Coursera

Languages

- **Native/ Professional Working Proficiency:** English, Bengali, Hindi
- **Beginner Proficiency:** French

Volunteering Activities

- **8th International Workshop on Event Sensing and Neuromorphic Engineering NeuroEng 2022**, Reviewer May 2022
- **Transactions on Machine Learning Research (TMLR)**, Volunteered the development of the OpenReview submission platform for TMLR journal Jan 2022

Invited Talks

- *Event-based Cameras: Present and Future Prospects*, Presented on January 2022, at Sprogen Services Private Limited, Pune, India

References

Available upon request